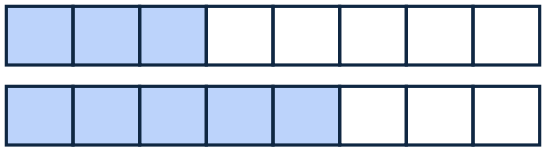
Comparing Fractions

Same whole? Then look at the pieces.

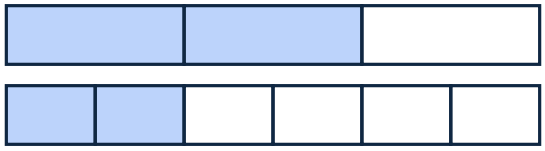
Same bottom number: same size pieces



$\frac{3}{8} < \frac{5}{8}$

More pieces of the same size is more.

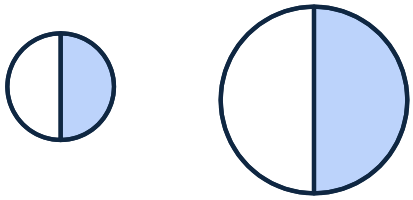
Same top number: look at the size



$\frac{2}{3} > \frac{2}{6}$

Same number of pieces, but thirds are bigger.

Only compare the same whole



Half of a small pizza is less than half of a big pizza.

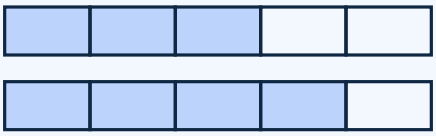
Compare fractions of the same whole.

Equal? The strips line up.

$\frac{2}{4} = \frac{1}{2}$

Use the fraction wall from Equal Fractions.

Your turn!



$\frac{3}{5}$ [<] $\frac{4}{5}$

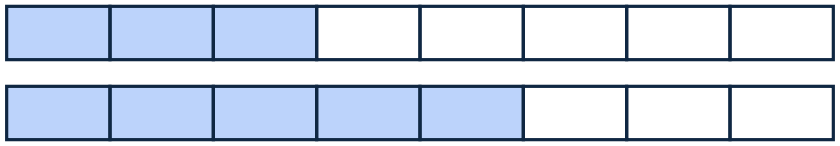
For teachers

Goal: compare two fractions of the same whole with <, > or = (3.NF.3d). Ask “Same size pieces? Then who has more pieces?” or “Same number of pieces? Then whose pieces are bigger?”

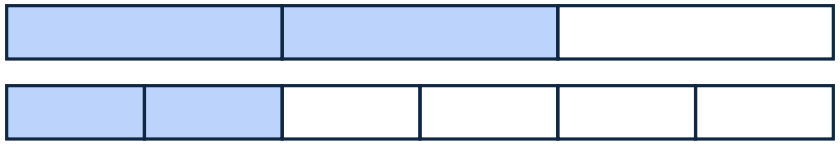
Comparing Fractions: Practice

Same whole. Write <, > or =.

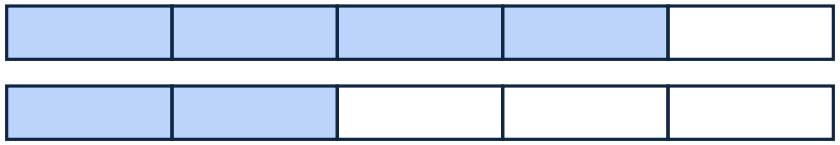
Part 1 Compare with strips Look at how much is shaded.

1) 

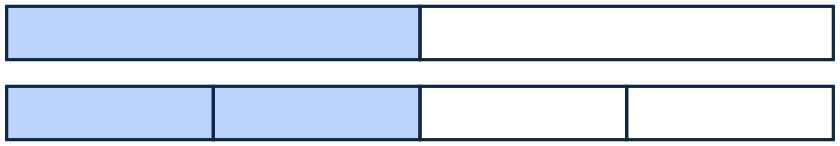
$\frac{3}{8} < \frac{5}{8}$

2) 

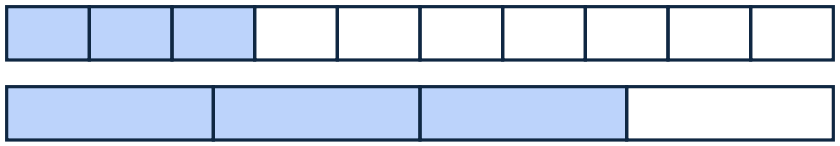
$\frac{2}{3} > \frac{2}{6}$

3) 

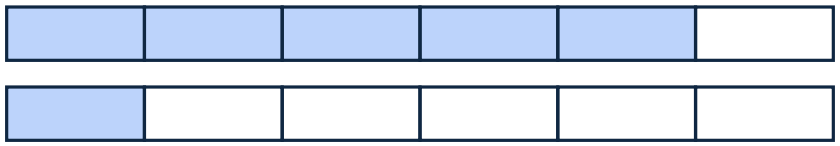
$\frac{4}{5} > \frac{2}{5}$

4) 

$\frac{1}{2} = \frac{2}{4}$

5) 

$\frac{3}{10} < \frac{3}{4}$

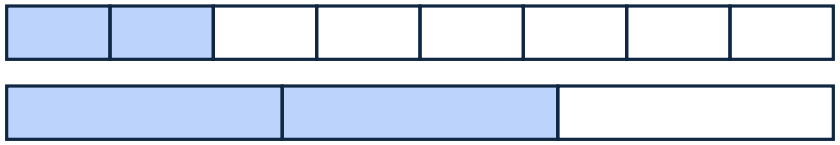
6) 

$\frac{5}{6} > \frac{1}{6}$

Comparing Fractions: Practice (continued)

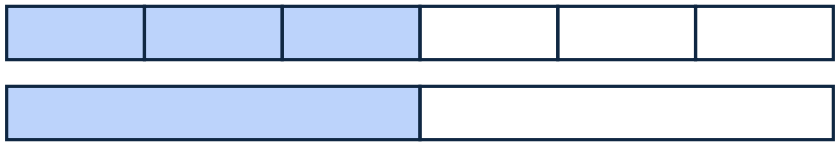
Part 1 Compare with strips (continued)

7)



$\frac{2}{8} < \frac{2}{3}$

8)



$\frac{3}{6} = \frac{1}{2}$

Part 2 Compare with shapes Same size shapes.

9)



$\frac{2}{4} < \frac{3}{4}$

10)



$\frac{1}{3} > \frac{1}{5}$

11)



$\frac{5}{6} > \frac{4}{6}$

12)



$\frac{2}{4} < \frac{2}{3}$

13)



$\frac{1}{2} = \frac{3}{6}$

14)



$\frac{3}{5} < \frac{4}{5}$

Part 3 Same bottom number No help this time. Same size pieces: more pieces is more.

15) $\frac{4}{6} > \frac{3}{6}$

16) $\frac{2}{5} < \frac{3}{5}$

17) $\frac{7}{8} > \frac{3}{8}$

Comparing Fractions: Practice (continued)

Part 3 Same bottom number (continued)

18) $\frac{1}{4} < \frac{3}{4}$

19) $\frac{6}{10} > \frac{4}{10}$

20) $\frac{2}{3} > \frac{1}{3}$

21) $\frac{3}{10} < \frac{9}{10}$

22) $\frac{5}{8} > \frac{1}{8}$

23) $\frac{1}{5} < \frac{4}{5}$

24) $\frac{5}{6} > \frac{2}{6}$

25) $\frac{2}{8} < \frac{6}{8}$

26) $\frac{7}{10} > \frac{2}{10}$

Part 4 Same top number No help this time. Same number of pieces: bigger pieces is more.

27) $\frac{2}{5} > \frac{2}{8}$

28) $\frac{3}{10} < \frac{3}{4}$

29) $\frac{3}{8} < \frac{3}{5}$

Comparing Fractions: Practice (continued)

Part 4 Same top number (continued)

30) $\frac{2}{3} > \frac{2}{4}$

31) $\frac{4}{6} > \frac{4}{10}$

32) $\frac{2}{10} < \frac{2}{6}$

33) $\frac{4}{8} < \frac{4}{5}$

34) $\frac{3}{4} > \frac{3}{6}$

35) $\frac{3}{6} < \frac{3}{5}$

36) $\frac{2}{4} > \frac{2}{10}$

37) $\frac{2}{8} < \frac{2}{5}$

38) $\frac{5}{8} > \frac{5}{10}$

Part 5 Unit fractions No help this time.

39) $\frac{1}{3} > \frac{1}{8}$

40) $\frac{1}{10} < \frac{1}{2}$

41) $\frac{1}{6} < \frac{1}{3}$

Comparing Fractions: Practice (continued)

Part 5 Unit fractions (continued)

42) $\frac{1}{4} > \frac{1}{5}$

43) $\frac{1}{2} > \frac{1}{4}$

44) $\frac{1}{8} < \frac{1}{6}$

45) $\frac{1}{3} < \frac{1}{2}$

46) $\frac{1}{5} > \frac{1}{10}$

47) $\frac{1}{6} > \frac{1}{8}$

Part 6 Equal or not? No help this time. Some are equal fractions.

48) $\frac{2}{4} = \frac{1}{2}$

49) $\frac{3}{8} < \frac{4}{8}$

50) $\frac{2}{6} = \frac{1}{3}$

51) $\frac{4}{5} > \frac{3}{5}$

52) $\frac{6}{8} = \frac{3}{4}$

53) $\frac{5}{10} = \frac{1}{2}$

Comparing Fractions: Practice (continued)

Part 6 Equal or not? (continued)

54) $\frac{2}{10}$ [=] $\frac{1}{5}$

55) $\frac{4}{8}$ [=] $\frac{2}{4}$

56) $\frac{5}{8}$ [>] $\frac{3}{8}$

Part 7 Circle the biggest No help this time.

57) **$\frac{5}{6}$** $\frac{2}{6}$ $\frac{3}{6}$

58) $\frac{1}{4}$ $\frac{1}{8}$ **$\frac{1}{2}$**

59) $\frac{3}{10}$ **$\frac{3}{4}$** $\frac{3}{5}$

60) **$\frac{7}{8}$** $\frac{4}{8}$ $\frac{2}{8}$

61) $\frac{2}{5}$ $\frac{2}{10}$ **$\frac{2}{3}$**

62) $\frac{1}{6}$ **$\frac{1}{3}$** $\frac{1}{10}$

Part 8 Same bottom Write <, > or =.

63) $\frac{5}{10}$ [<] $\frac{8}{10}$

64) $\frac{3}{4}$ [>] $\frac{1}{4}$

65) $\frac{4}{6}$ [>] $\frac{2}{6}$

Comparing Fractions: Practice (continued)

Part 8 Same bottom (continued)

66) $\frac{1}{6} < \frac{2}{6}$

67) $\frac{5}{10} < \frac{9}{10}$

68) $\frac{2}{6} < \frac{5}{6}$

69) $\frac{3}{6} < \frac{5}{6}$

70) $\frac{1}{3} < \frac{2}{3}$

71) $\frac{4}{12} < \frac{11}{12}$

72) $\frac{7}{8} > \frac{1}{8}$

73) $\frac{8}{10} > \frac{1}{10}$

74) $\frac{3}{12} < \frac{5}{12}$

75) $\frac{5}{6} > \frac{1}{6}$

76) $\frac{4}{8} > \frac{2}{8}$

77) $\frac{5}{8} > \frac{2}{8}$

Comparing Fractions: Practice (continued)

Part 8 Same bottom (continued)

78) $\frac{1}{12} < \frac{2}{12}$

79) $\frac{8}{12} < \frac{9}{12}$

80) $\frac{6}{12} > \frac{2}{12}$

81) $\frac{3}{12} < \frac{4}{12}$

82) $\frac{1}{10} < \frac{9}{10}$

83) $\frac{5}{12} < \frac{10}{12}$

84) $\frac{7}{10} > \frac{6}{10}$

85) $\frac{1}{6} < \frac{3}{6}$

86) $\frac{1}{6} < \frac{4}{6}$

Part 9 Same top

87) $\frac{3}{10} < \frac{3}{6}$

88) $\frac{2}{4} > \frac{2}{12}$

89) $\frac{1}{10} < \frac{1}{8}$

Comparing Fractions: Practice (continued)

Part 9 Same top (continued)

90) $\frac{3}{4} > \frac{3}{12}$

91) $\frac{2}{10} < \frac{2}{4}$

92) $\frac{3}{10} > \frac{3}{12}$

93) $\frac{2}{12} < \frac{2}{4}$

94) $\frac{1}{12} < \frac{1}{8}$

95) $\frac{2}{8} < \frac{2}{6}$

96) $\frac{1}{8} > \frac{1}{12}$

97) $\frac{3}{6} > \frac{3}{10}$

98) $\frac{1}{12} < \frac{1}{10}$

99) $\frac{3}{4} > \frac{3}{10}$

100) $\frac{1}{8} > \frac{1}{10}$

101) $\frac{3}{12} < \frac{3}{4}$

Comparing Fractions: Practice (continued)

Part 9 Same top (continued)

102) $\frac{2}{6}$ > $\frac{2}{12}$

103) $\frac{3}{12}$ < $\frac{3}{8}$

104) $\frac{3}{6}$ > $\frac{3}{12}$

Part 10 Remember? Problems from earlier packets.

105) $\frac{4}{5} = \frac{8}{10}$

106) $\frac{12}{3} = 4$

107) Round 384 to the nearest hundred: 400

108) 3 groups of 8 = 24

109) 9 m by 4 m: area = 36 sq m

110) two eighths = $\frac{2}{8}$

Part 11 Word problems Draw strips to compare.

111) Kim ate 3/8 of a pizza. Leo ate 5/8 of the same pizza. Who ate more?

Number sentence 3/8 < 5/8

Answer Leo

Draw or show your work

112) Ana ran 2/3 of a mile. Ben ran 2/6 of a mile. Who ran farther?

Number sentence 2/3 > 2/6

Answer Ana

Draw or show your work

113) A red ribbon is 1/4 of a yard. A blue ribbon is 1/2 of a yard. Which one is shorter?

Number sentence 1/4 < 1/2

Answer red ribbon

Draw or show your work

Comparing Fractions: Practice (continued)

Part 11 Word problems (continued)

114) Mia's glass is 4/6 full. Sam's same-size glass is 2/3 full. Who has more juice?

Number sentence 4/6 = 2/3

Answer same

Draw or show your work

115) Challenge: Jo says half of a small pizza is as much as half of a big pizza. Is Jo right?

Number sentence 1/2 of different wholes

Answer no

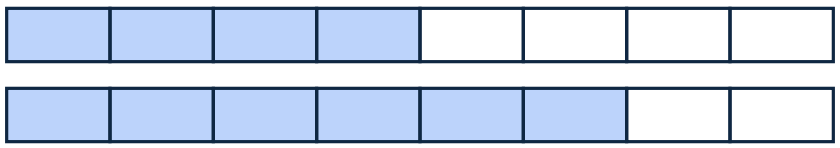
Draw or show your work

Check-In: Comparing Fractions

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Compare with strips

1) 

$\frac{4}{8} < \frac{6}{8}$

Part 2 Same bottom number Write <, > or =.

2) $\frac{5}{8} < \frac{6}{8}$

3) $\frac{2}{8} < \frac{7}{8}$

4) $\frac{5}{10} > \frac{2}{10}$

Part 3 Same top number Write <, > or =.

5) $\frac{2}{8} < \frac{2}{4}$

6) $\frac{3}{10} < \frac{3}{8}$

7) $\frac{2}{4} > \frac{2}{8}$

Part 4 Circle the biggest

8) $\frac{2}{8}$ $\frac{3}{8}$ $\frac{4}{8}$

9) $\frac{1}{5}$ $\frac{4}{5}$ $\frac{2}{5}$